

Friedrichs Aharonov–Bohm Dynamics: Complete Scattering and the Forward Distribution

HCS-C383 theorem and reproducibility package

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Abstract

We carry a fixed Friedrichs domain and flux sign through the complete spectral and scattering theory of the punctured-plane magnetic Laplacian. Incoming and outgoing Bessel asymptotics determine the wave operators and the exact phases $S_m = \exp\{i\pi(|m| - |m - \beta|)\}$. All angular channels are included by an onto Hankel transform and a dense-packet strong-limit argument. Abel summation then gives the full angular scattering distribution: a forward delta, a principal-value cotangent, and a constant imaginary term. The familiar differential cross section is recovered only away from the forward ray. This round closes complete source scattering without silently replacing its distributional kernel by an ordinary function.

中文摘要

本文在同一弗里德里希斯边界条件下闭合穿孔平面磁性算子的完整散射。
径向贝塞尔函数的大参数渐近分别固定入射与出射相位，汉克尔变换的满射性与稠密波包极限给出完整波算子。
角向散射相位按正负通道分成两个常值半空间，但通道零点的归属不能省略。
通过阿贝尔求和，散射核同时包含前向狄拉克分布、余切主值与常数虚部；
缺少前向项的普通函数并不表示完整酉散射算子。
熟知的微分截面仅在离开前向射线时有函数意义，不能把分布平方当成前向截面。
这一轮把谱域、波算子相位、全通道完备性与分布归一化连接成一个可核查的散射终点。

Keywords: Aharonov–Bohm scattering; wave operators; angular channels; Abel summation; forward distribution; cross section

关键词：阿哈罗诺夫玻姆散射；波算子；角通道；阿贝尔求和；前向分布；散射截面

1 One operator and one boundary condition

On $\mathcal{H} = L^2(\mathbb{R}^2, r dr d\theta)$, set $h = 2M = 1$ and freeze the nonnegative form, initially on smooth functions compactly supported away from zero,

$$q_\beta[f] = \int_0^\infty \int_0^{2\pi} \left(|\partial_r f|^2 + \frac{|(-i\partial_\theta - \beta)f|^2}{r^2} \right) r d\theta dr. \quad (1)$$

Its closed Friedrichs extension defines the self-adjoint H_β . The main flux representative is $0 \leq \beta < 1$; integer translates will only be used for gauge comparisons. No spin, contact interaction, alternate self-adjoint extension, finite-radius flux tube, or confining field enters. This choice is material: an unqualified singular differential expression does not fix its quantum dynamics.

Let $e_m(\theta) = (2\pi)^{-1/2} e^{im\theta}$ and $\nu_m = |m - \beta|$. Angular Parseval gives the orthogonal form sum

$$q_\beta = \bigoplus_{m \in \mathbb{Z}} q_{\nu_m}, \quad q_\nu[u] = \int_0^\infty \left(|u'|^2 + \frac{\nu^2}{r^2} |u|^2 \right) r dr. \quad (2)$$

Finite angular sums and smooth compact radial approximants form a core of this sum: Fourier truncation converges in both norm and form energy, and the radial closure then approximates each of finitely many components.

Theorem 1 (Domain and complete spectral representation). *The Friedrichs operator decomposes as*

$$H_\beta = \bigoplus_{m \in \mathbb{Z}} h_{\nu_m}, \quad h_\nu = -\partial_r^2 - r^{-1} \partial_r + \nu^2 r^{-2}. \quad (3)$$

The order- ν Hankel map

$$(\mathcal{J}_\nu u)(k) = \int_0^\infty J_\nu(kr) u(r) r dr \quad (4)$$

is unitary from $L^2(r dr)$ onto $L^2(k dk)$ and transforms h_ν to multiplication by k^2 . In particular,

$$D(h_\nu) = \{u : k^2 \int_\nu u \in L^2(k dk)\}. \quad (5)$$

The full operator has purely absolutely continuous spectrum $[0, \infty)$, with no point or singular continuous spectrum.

Proof. The form-core argument proves (3). For $0 < \nu < 1$, the two zero-energy branches are r^ν and $r^{-\nu}$. The latter has divergent form energy and its coefficient is removed in the Friedrichs realization. For $\nu = 0$, the logarithmic branch is excluded but the regular constant branch is allowed. Orders $\nu \geq 1$ are limit point at zero, and infinity is limit point in every channel. Thus these are precisely the regular Bessel realizations diagonalized by the Hankel Plancherel and inversion theorem [3]. The transform is onto and gives (5); it is not a finite list of formal eigenfunctions. Multiplication by k^2 under the non-atomic measure $k dk$ has the claimed spectral types, and its countable orthogonal sum preserves them. $\square$

Lemma 1 (The zero-flux puncture). H_0 is the ordinary free plane Laplacian.

Proof. For $0 < \epsilon < 1$, take a radial cutoff equal to zero at $r \leq \epsilon^2$, equal to one at $r \geq \epsilon$, and linear in $\log r$ between those radii. Its Dirichlet energy is $2\pi/|\log \epsilon|$. Multiplying a fixed smooth function by this cutoff changes its L^2 norm and Dirichlet energy by terms tending to zero. Hence the puncture has zero H^1 capacity and the closed zero-flux form equals the full-plane free form. $\square$

The original physical effect belongs to Aharonov and Bohm [1]; the rigorous spectral and scattering treatment for general extension parameters is reviewed by Pankrashkin and Richard [2]. Their magnetic-potential sign corresponds to orders $|m + \alpha|$. Angular reflection, or $\alpha = -\beta$, translates to (3). The present formulas use one sign throughout. Reconstructing and testing this classical theory is the purpose of this package; classical formula discovery is not claimed.

2 Complete wave operators in the fixed sign convention

Put $\mu_m = |m|$ and $a_m = \pi(\mu_m - \nu_m)/2$. Define the free comparison by

$$W_{\pm} = \text{s-lim}_{t \rightarrow \pm\infty} e^{itH_{\beta}} e^{-itH_0}, \quad S = W_{+}^{*} W_{-}. \quad (6)$$

Theorem 2 (Wave operators and all scattering phases). *The limits in (6) exist and are onto. Their channel restrictions and the on-shell angular scattering phases are*

$$W_{\pm, m} = e^{\mp i a_m} \mathcal{J}_{\nu_m} \mathcal{J}_{\mu_m}, \quad S_m = e^{i\pi(|m| - |m - \beta|)}. \quad (7)$$

For $0 < \beta < 1$ this is $e^{i\pi\beta}$ on $m \geq 1$ and $e^{-i\pi\beta}$ on $m \leq 0$.

Proof. The real-order large-argument expansion is

$$J_{\nu}(kr) = \left(\frac{2}{\pi kr} \right)^{1/2} \cos(kr - \pi\nu/2 - \pi/4) + O((kr)^{-3/2}). \quad (8)$$

The incoming radial coefficient has phase $e^{i\pi\nu/2}$ and the outgoing one phase $e^{-i\pi\nu/2}$. Matching them to the free order μ_m gives e^{ia_m} for W_{-} and e^{-ia_m} for W_{+} . The rigorous dense-packet argument takes smooth spectral support in a compact subset of $k > 0$. Stationary phase puts the incoming packet at $r \approx 2|t|/k$ for negative time and the outgoing packet there for positive time. The remainder in (8) tends to zero in radial L^2 in that region, while integration by parts controls the nonstationary complement. This proves the channel limits on a dense set. The wave-operator theorem in [2, Lemma 7 and the original AB channel calculation] supplies the rigorous existence and completeness statement for exactly this regular extension. The explicit phase matching fixes its sign in our convention. Uniform unitary bounds extend the channel limits to all radial vectors; finite angular sums extend them to $\mathcal{H}$. Every Hankel product is onto, so their direct sum is onto. Multiplication of $W_{+, m}^{*}$ and $W_{-, m}$ gives the last formula. Splitting at $m = 0$ proves the two constant phases. $\square$

3 The forward distribution is part of scattering

Let P_{+} select $m \geq 1$ and P_{-} select $m \leq 0$ in $L^2(S^1, d\theta)$. Then $S = e^{i\pi\beta} P_{+} + e^{-i\pi\beta} P_{-}$. Write $\delta_{2\pi}$ for the periodic delta with integral one, and let θ be outgoing angle minus incoming angle.

Theorem 3 (Full kernel and away-forward cross section). *The convolution kernel relative to $d\theta$ is*

$$K_S(\theta) = \cos(\pi\beta) \delta_{2\pi}(\theta) - \frac{\sin(\pi\beta)}{2\pi} \text{PV} \cot(\theta/2) - \frac{i \sin(\pi\beta)}{2\pi}. \quad (9)$$

Under the amplitude convention $f = (2\pi i k)^{-1/2} \sum_m (S_m - 1) e^{im\theta}$, with $\sqrt{i} = e^{i\pi/4}$, away from $\theta = 0$ modulo 2π one has

$$f = -\frac{\sin(\pi\beta)}{\sqrt{2\pi i k}} (\cot(\theta/2) + i), \quad \frac{d\sigma}{d\theta} = \frac{\sin^2(\pi\beta)}{2\pi k \sin^2(\theta/2)}. \quad (10)$$

Proof. Introduce Abel weights $\rho^{|m|}$, $0 < \rho < 1$, in the geometric series. Their distributional limits are

$$\sum_{m \geq 1} e^{im\theta} = \pi \delta_{2\pi} - \frac{1}{2} + \frac{i}{2} \text{PV} \cot(\theta/2), \quad (11)$$

$$\sum_{m \leq 0} e^{im\theta} = \pi \delta_{2\pi} + \frac{1}{2} - \frac{i}{2} \text{PV} \cot(\theta/2). \quad (12)$$

Multiplying by the two channel phases and dividing by 2π gives (9). Subtracting the identity kernel retains $(\cos \pi\beta - 1) \delta_{2\pi}$. Off its support the amplitude equals the function in (10); $|\cot(\theta/2) + i|^2 = \csc^2(\theta/2)$ gives the cross section. No square of a distribution is taken. $\square$

For nonzero flux in the chosen interval the integrated away-forward cross section diverges. The two angular projectors nevertheless show S is exactly unitary. These statements concern different objects and are consistent.

References

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- [3] NIST Digital Library of Mathematical Functions, Section 10.17 and Section 10.22, Hankel transforms and Weber’s exponential integral, accessed 5 September 2026.

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